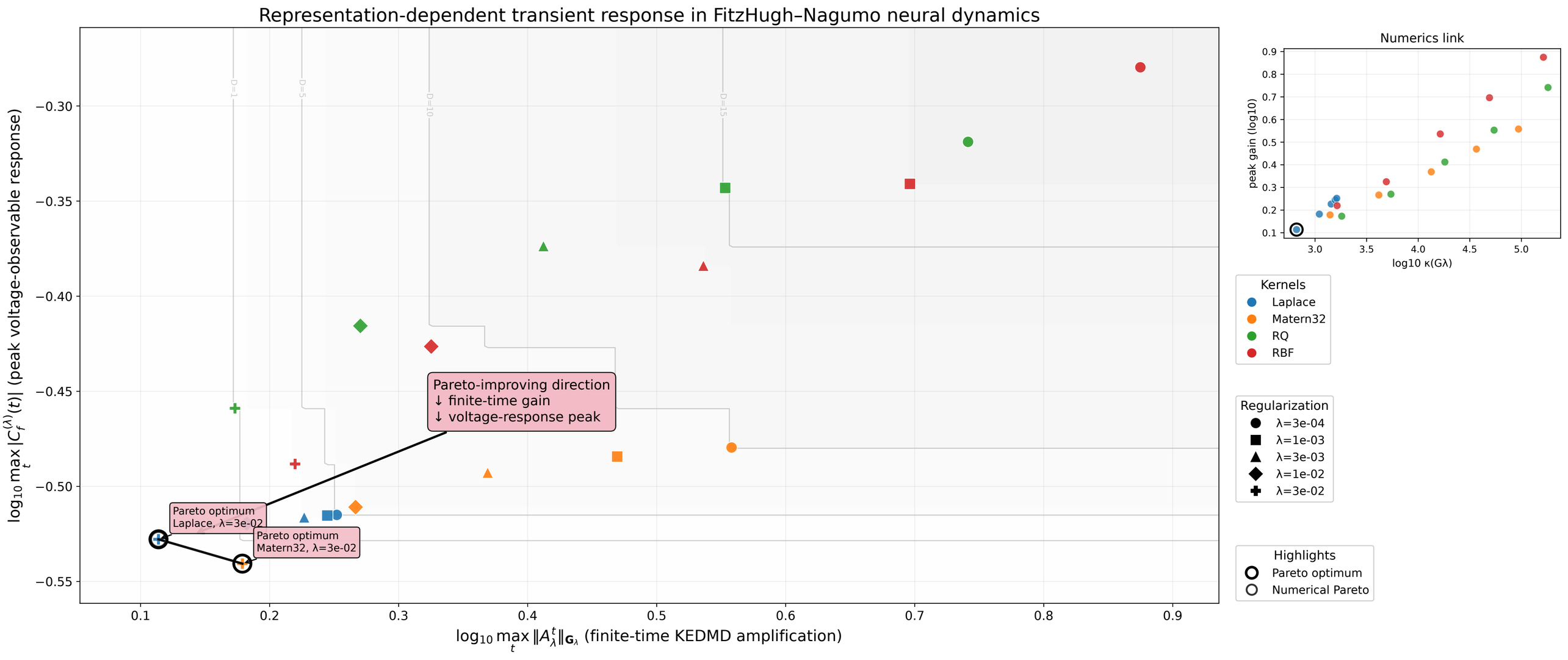

Representation-dependent transient response in FitzHugh–Nagumo neural dynamics
$\log_{10} \max_t |C_f^{(\lambda)}(t)|$ (peak voltage-observable response)
$\log_{10} \max_t \|A_\lambda^t\|_{\mathbf{G}_\lambda}$ (finite-time KEDMD amplification)
D=1
D=5
D=10
D=15
Pareto-improving direction
↓ finite-time gain
↓ voltage-response peak
Pareto optimum
Laplace, λ=3e-02
Pareto optimum
Matern32, λ=3e-02
Numerics link
peak gain (log10)
log10 κ(Gλ)
Kernels
Laplace
Matern32
RQ
RBF
Regularization
λ=3e-04
λ=1e-03
λ=3e-03
λ=1e-02
λ=3e-02
Highlights
Pareto optimum
Numerical Pareto